

IDENTIFICATION

PRODUCT CODE: MAINDEC-X8-DIKAD-B-D
PRODUCT NAME: DEC/X8 MODULE "NOTFUN"
NON-FUNCTIONAL IOT TEST
DATE CREATED: MAY 18, 1972
MAINTAINER: DIAGNOSTIC GROUP
AUTHOR: JOHN VROBEL

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MAYNARD, MASS, 01754

1, MODULE DESCRIPTION

"NOTFUN" IS A DEC/X8 SOFTWARE MODULE WHICH VERIFIES THAT ALL NON-FUNCTIONAL IOT'S WITHIN A GIVEN SYSTEM DO NOT AFFECT THAT SYSTEM WHEN EXECUTED,

THE METHOD USED IS TO EXECUTE ALL IOT'S NOT INCLUDED IN A USER SUPPLIED LIST OF FUNCTIONAL IOT'S AND VERIFYING DIRECTLY THAT THE AC IS UNEFFECTED AND THAT NO SKIP OCCURS, THE DEC/X8 MONITOR AND/OR OTHER EXERCISER MODULES SHOULD DETECT ANY MORE SUBTLE INTERACTIVE PROBLEMS,

2, REQUIREMENTS

- 1, PROCESSORS: PDP-8,8/I,8/L,8/E,8/M AND PDP-12
- 2, OPTIONS: NONE
- 3, SPECIAL: NONE

3, RESTRICTIONS

NONE

4, OPERATING INFORMATION

4,1 SPECIAL CONSIDERATIONS

NONE

4,2 BUILDING

- 1, JOB TYPE: BACKGROUND
- 2, PRIORITY: MUST HAVE LOWER PRIORITY THAN ALL INTERRUPT JOBS; OTHERWISE UNIMPORTANT,
- 3, JOB SLOTS: ANY EXISTENT JOB SLOT; 2 PAGES REQUIRED,

4,3 INITIALIZING

AFTER "NOTFUN" IS PRINTED, TYPE "0" FOR A PDP-8/E OR 8/M, OR A "1" FOR A PDP-8,8/I,8/L OR PDP-12,

THEN AN OCTAL LIST STARTING AT "01" WILL BE PRINTED, FOLLOWING THE NUMBERED ITEMS IN THE LIST TYPE ALL EXISTING DEVICE CODES (EXCEPT 00) IN THE SYSTEM UNDER TEST, IF THE SYSTEM HAS EXTENDED MEMORY, DEVICE CODES 20 THROUGH 27 MUST BE SPECIFIED, WHEN COMPLETE TYPE "00" TO SIGNIFY THE END OF INPUT,

BELOW IS A TYPICAL EXAMPLE!

```
NOTFUN  0      (PDP-8/E)
01  03      (KEYBOARD)
02  04      (TELEPRINTER)
03  76      (DECTAPE)
04  77      (DECTAPE)
05  66      (LP08/LE8)
06  01      (READER)
07  02      (PUNCH)
10  20      (EXTENDED MEMORY DEVICE CODES 20-27)
11  21
12  22
13  23
14  24
15  25
16  26
17  27
20  00      (END OF INPUTS)
```

4.4 DEVICE SETUP

NONE

4.5 RUNNING

```
1,  CNTR:  UPDATED AFTER EACH IOT IS TESTED.
2,  SR10:  NO EFFECT
3,  SR11:  NO EFFECT
```

5. ERROR INFORMATION

ALL ERRORS ARE REPORTED AS "SIAT ERR" AND ARE IN A WAY CONSIDERED FATAL. THE JOB IS THEREFORE AUTOMATICALLY KILLED AFTER ANY ERROR IS DETECTED.

5.1 ERROR SYMBOL DEFINITIONS

```
CODE:  0000  SETUP ERROR, THE USER HAS SPECIFIED THAT ALL DEVICE
          CODES ARE USED IN THE SYSTEM UNDER TEST OR THE MODULE
          HAS NEVER BEEN INITIALIZED.
          7776  ILLEGAL CHANGE IN THE AC.
          7777  ILLEGAL SKIP
```

```
SAI  IOT EXECUTED
SBI  GOOD AC
SCI  AQ AFTER IOT WAS EXECUTED
```

6. LISTING (ATTACHED)

/DEC/X8 EXTERNAL SYMBOL TABLE "EXTSYN"
 /FOR USE IN ASSEMBLING DEC/X8 SOFTWARE MODULES;
 /COPYRIGHT 1972, DIGITAL EQUIPMENT CORP., MAYNARD, MASS.
 XLIST
 PAUSE

PAL18 V141 18-MAY-72 13148 PAGE 2

/MAINDEC-X8-DIKAD-B-L "DEC/X8" NOTFUN
 /EXECUTES ALL IOTS NOT USED ON THE SYSTEM,
 /COPYRIGHT 1972, DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS,
 /THIS MODULE OPERATIONAL ON PDP-8,8/1,8/L,8/E AND PDP-12,
 /PRG: JOHN VROBEL (X3621)

/BUILDER INSTRUCTIONS

/1, PRIORITY: BACKGROUND JOB
 /2, JOB SLOT: 2 PAGES REQUIRED, ANY EXISTENT JOB SLOT;

/INITIALIZER INSTRUCTIONS

/1, AFTER TITLE IS PRINTED, TYPE IN A B FOR PDP8/E OR A 1
 /FOR PDP8, PDP8/I OR PDP8/L,
 /2, THEN AN OCTAL LIST WILL BE PRINTED, STARTING WITH 01,
 /FOR INPUT OF USED IOT CODES, TYPICAL LIST SHOWN BELOW;
 /3, AFTER ALL USED IOT CODES ARE TYPED IN, TYPE IN A 00
 /TO SIGNIFY LAST ENTRY;

```

/      NOTFUN 1
/      01 03
/      02 04
/      03 05
/      04 06
/      05 07
/      06 13
/      07 02
/
/      /SIGNIFIES LAST ENTRY
  
```

```

0200 0200 0200
0201 1617 JOB, 0
0202 2406 TEXT1, TEXT "NOTFUN"
0203 2516
0204 0000
0205 0411 TEXT "DIKAD=B"
0206 1301
0207 0435
0210 0200
/
0211 0000 HONEDF, 0
0212 7402 HLT/ODF
0213 5611 JMP I HONEDF
/
0214 0202 INTACK, CIF 00
0215 4426 JMS I IHRET
  
```

```

0216 7777      -1
0217 7777      KILL, -1
0220 7777      KILLED, -1
0221 0000      CNTR, 0
/
0222 0000      ERROR, 0
0223 3234      DCA ,+11
0224 7604      LAS
0225 0073      AND Z K4
0226 7440      SZA
0227 3217      DCA KILL
0230 4211      JMS HOMEOP
0231 6002      IOF
0232 6202      CIF 00
0233 4461      JMS I ERRP
0234 0000      0
0235 5622      JMP I ERROR
0236 0000      CODE, 0
0237 7775      -3
0240 0000      ERRSA, 0
0241 0000      ERRSB, 0
0242 0000      ERRSC, 0

```

/END MODULE TABLE
/INITIALIZATION ROUTINE

```

0243 1377      INIT, TAD (TEXT1
0244 3246      DCA ,+2
0245 4444      MESSAGE
0246 0000      0
0247 4455      SPACE2
0250 4442      ONEOCI
0251 5247      JMP ,+2
0252 3354      DCA KIND
0253 7340      CLA CLL CHA
0254 3353      DCA SAVAC1
0255 3776      DCA STRBUF
0256 1355      TAD A7701
0257 3356      DCA INCNT
0260 1361      TAD ABUF
0261 3010      AUTO1, DCA 10
0262 4454      RECEV, CRLF
0263 1356      TAD INCNT
0264 4452      PRNT2
0265 4455      SPACE2
0266 4441      TWOOCI
0267 5262      JMP RECEV
0270 7450      SNA
0271 5300      JMP ,+7
0272 7104      RAL CLL
0273 7006      RTL
0274 3410      AUTO2, DCA I 10
0275 2356      ISZ INCNT
0276 5262      JMP RECEV

```

/WAIT FOR TTY INPUT
/ERROR ON INPUT
/SAVE JOB

/PRINT TWO OCTAL NUMBERS
/THEN TWO SPACES
/RECEIVE TWO OCTAL NUMBERS
/ERROR START OVER
/HAS IT END OF LIST
/YES

/NO, STORE 10T CODE
/MAKE SURE OPERATOR DOESN'T GOOF
/OK, CONTINUE

```

0277 3353      DCA SAVAC1
0300 3410      AUTO3, DCA I 10
0301 5020      INITEX
/
0302 3242      SKPERR, DCA ERRSC
0303 7340      CLA CLL CHA
0304 5236      DCA CODE
0305 1141      TAD M3
0306 3237      DCA ERRSA -1
0307 5317      JMP PNTR1
/
0310 7344      ACERR, CLA CLL CHA RAL
0311 3236      DCA CODE
0312 1141      TAD M3
0313 3237      DCA ERRSA -1
0314 5317      JMP PNTR1
/
0315 3236      SETERR, DCA CODE
0316 3237      DCA ERRSA -1
/
0317 4222      PNTR1, JMS ERROR
0320 7340      CLA CLL CHA
0321 6002      IOF
0322 3220      DCA KILLED
0323 5004      SERVEX
/
0324 0000      OUT, 0
0325 2221      ISZ CNTR
0326 7000      NOP
0327 1217      TAD KILL
0330 7450      SNA
0331 5334      JMP ,+3
0332 3220      DCA KILLED
0333 5004      SERVEX
0334 1324      TAD OUT
0335 5004      SERVEX
/
0336 3221      RUN, DCA CNTR
0337 1353      TAD SAVAC1
0340 7650      SNA CLA
0341 5315      JMP SETERR
0342 1357      BGN, TAD KIOT
0343 3775      DCA TSTIOT
0344 5774      JMP LPSK
0345 2775      UPIOT, ISZ TSTIOT
0346 1775      TAD TSTIOT
0347 1360      TAD M7000
0350 7650      SNA CLA
0351 5342      JMP BGN
0352 5774      JMP LPSK

```

/CLEAR MONITOR PASS COUNTER

/HAS IT A SETUP ERROR
/YES KILL JOB
/00010

/HAS IT LAST 10T

```

0353 0000      SAVAC1, 0000
0354 0000      KIND, 0
0355 7701      A7701, 7701
0356 0000      INCNT, 0

```

/PRESET TO GIVE ERROR IF NOT INITIALIZED

```

0357 0010 K10T, 0010
0360 1000 M7000, -7000
/
0361 0,
/
0361 0461 ABUF, STRBUF -1
/
0374 0400
0375 0440
0376 0462
0377 0201
0400 0400
/
0400 1257 LPSK, TAD B7700
0401 3260 DCA LOOP
0402 1362 TAD BBUF /COUNTER FOR EACH IOT
0403 3010 AUT04, DCA 10 /START OF BUFFER -1
0404 1240 TAD TST10T
0405 0256 AND K0770 /MASK 3-8
0406 3255 DCA SAVAC2
0407 1410 COMNX, TAD 1 10
0410 7450 SNA /HAS IT AND OF BUFFER
0411 9217 JMP EX10T /YES EXECUTE THE IOT
0412 7041 CIA
0413 1255 TAD SAVAC2
0414 7650 SNA CLA
0415 9777 JMP UP10T
0416 9207 JMP COMNX
/
0417 1240 EX10T, TAD TST10T
0420 3776 DCA ERRSA /STORE FOR TYPE OUT
0421 1775 TAD K10 /TYPE OF COMPUTER
0422 7650 SNA CLA
0423 9230 JMP ,+5
0424 1240 TAD TST10T
0425 0075 AND K7
0426 7650 SNA CLA /HAS IT IOT 0
0427 9777 JMP UP10T /GO TO NEXT
0430 1661 REEX, TAD 1 RANDAT /GENERATE RANDOM NUMBER
0431 1774 TAD ERRSB
0432 2261 ISZ RANDAT
0433 7012 RTR
0434 7420 SNL
0435 1000 TAD 0
0436 3774 DCA ERRSB
0437 1774 TAD ERRSB
0440 7402 TST10T, HLT
0441 7410 SKP /DIDN'T SKIP
0442 9773 JMP SKPERR /SKIP ERROR
0443 3772 DCA ERRSC /STORE AC FOR TYPE OUT
0444 1772 TAD ERRSC
0445 7041 CIA
0446 1774 TAD ERRSB
0447 7640 SZA CLA /COMPARE TO INPUT AC
0450 9771 JMP ACERR /AC ERROR

```

```

0451 2260 ISZ LOOP
0452 5230 JMP REEX /EXECUTE SAME IOT
0453 4770 JMS OUT
0454 9777 JMP UP10T /GO TO NEXT IOT
/
0455 0000 SAVAC2, 0
0456 0770 K0770, 0770
0457 7700 07700, 7700
0460 0000 LOOP, 0000
0461 2000 RANDAT, 2000
/
0462 STRBUF,
/
0562 0, +100
/
0562 0461 BBUF, STRBUF -1
/
/LOADER CALL
/
0570 0324
0571 0310
0572 0242
0573 0302
0574 0241
0575 0354
0576 0240
0577 0345

```

```

0000
0100
0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11000000 00001111
0400 11111111 11111111 11111111 11111111 11111111 11111111 11000000 00000000
0500 00000000 00000000 00000000 00000000 00000000 00000000 00100000 11111111

```

```

0600
0700

```

```

1000
1100

```

```

1200
1300

```

```

1400
1500

```

```

1600
1700

```

```

2000
2100

```

```

2200
2300

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2400
2500

```

```

2600
2700

```

```

3000
3100

```

```

3200
3300

```

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3400
3500

```

```

3600
3700

```

```

4000
4100

```

```

4200
4300

```

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4400
4500

```

```

4600
4700

```

```

5000
5100

```

```

5200
5300

```

```

5400
5500

```

```

5600
5700

```

```

6000
6100

```

```

6200
6300

```

```

6400
6500

```

```

6600
6700

```

```

7000
7100

```

```

7200
7300

```

```

7400
7500

```

```

7600
7700

```

0001	FIELD 1
0600 0243	INIT
0601 0336	RUN
0602 0000	0
0603 7773	-5
0604 0261	AUTO1
0605 0274	AUTO2
0606 0300	AUTO3
0607 0403	AUTO4
0610 0407	COMNX
0611 0000	0
0612 0000	0
0613 0000	0

5

0000
0100
0200
0300
0400
0500

0600 1111111 11110000 00000000 00000000 00000000 00000000 00000000 *0000000
0700 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000

1000
1100
1200
1300
1400
1500
1600
1700

2000
2100
2200
2300
2400
2500
2600
2700

3000
3100
3200
3300
3400
3500
3600
3700

4800
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

A7701	0355	K280	0114	M43	0132
ABUF	0361	K272	0115	M5	0137
ACERR	0310	K277	0116	M7	0136
ASBUFF	4460	K3	0072	M7000	0360
ASBUFF	0060	K30	0103	MESSAGE	4444
AUTO	0017	K301	0117	MESSAGE	0044
AUTO1	0261	K32	0067	MUL20P	0065
AUTO2	0274	K323	0120	ONEOCP	0042
AUTO3	0300	K4	0073	ONEOCT	4442
AUTO4	0403	K40	0104	OUT	0324
B7700	0457	K400	0121	PNTR1	0317
BBUF	0562	K5	0074	PRNT1	4461
BGN	0342	K5200	0123	PRNT1P	0051
CNTR	0221	K540	0124	PRNT2	4452
CODE	0236	K5402	0003	PRNT2P	2052
COMNX	0407	K64	0070	PRNT4	4453
CRLF	4454	K7	0075	PRNT4P	0053
CRLF	0054	K70	0105	RANDAT	0461
ERROR	0222	K7510	0125	RECEV	0262
ERRP	0061	K7520	0126	REEX	0430
ERRSA	0240	K7540	0127	RLBUFF	4457
ERRSB	0241	K7600	0131	RLBUFF	0057
ERRSC	0242	K77	0106	RUN	0336
EXINIT	0020	K7735	0132	SAVAC1	0353
EXIOT	0417	K7740	0133	SAVAC2	0455
EXSERV	0004	K7750	0134	SERVEX	5004
EXTMEM	0161	K7760	0135	SETERR	0315
FOROCP	0043	K7771	0136	SKPERR	0302
FOROCT	4443	K7773	0137	SPACE2	4455
HOMEDF	0211	K7774	0140	SPACEP	0055
IHRETP	0026	K7775	0141	STRBUF	0462
INCNT	0356	KCDF	0064	TEXT1	0201
INIT	0243	KCIF	0005	YSTIOT	0440
INITEX	5020	KCIFDF	0020	THOOC	0041
INTACK	0214	KILL	0217	THOCT	4441
IOFNSP	0056	KILLED	0220	TYPE	4450
JOB	0200	KIND	0354	TYPEP	0050
K0	0066	KIOF	0004	UPIOT	0345
K0770	0456	KIOT	0357		
K10	0076	LISN	4440		
K100	0107	LISNP	0040		
K11	0077	LOOP	0460		
K116	0071	LPRK	0400		
K13	0100	M20	0135		
K17	0101	M200	0131		
K177	0130	M240	0127		
K20	0102	M260	0126		
K200	0110	M270	0125		
K2000	0122	M3	0141		
K212	0111	M30	0134		
K215	0112	M4	0140		
K240	0113	M40	0133		

ERRORS DETECTED: 0

LINKS GENERATED: 20

RUN-TIME: 4 SECONDS

2K CORE USED